

Software Requirements Specification

Modern Residential Electricity Bill Calculator

COSC 21052 / BECS 21213 – Software Engineering

Department of Statistics & Computer Science

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1. Introduction

The Modern Residential Electricity Bill Calculator is a Java-based desktop application developed to automate the computation of monthly electricity bills for domestic consumers in Sri Lanka. The system addresses the complexity introduced by the multi-tier slab tariff structure and the Time-of-Use (TOU) rate scheme effective from October 2025, as published by the Public Utilities Commission of Sri Lanka (PUCSL). It also incorporates solar energy export credits through Net Metering, reflecting the growing adoption of rooftop solar installations.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete and structured description of the requirements for the Modern Residential Electricity Bill Calculator. It defines the system's functional capabilities, non-functional quality attributes, external interfaces, and constraints. This document serves as a formal agreement between the development team and stakeholders, ensuring a shared understanding of what the system shall deliver.

1.2 Intended Audience

This document is intended for the following audiences:

- Software Developers – to guide implementation and testing of the system.
- Course Instructors and Evaluators – to assess the completeness and quality of the requirements analysis.
- Project Team Members – to serve as a reference for development decisions and future maintenance.
- End Users (Domestic Consumers) – to understand the capabilities and limitations of the system.

1.3 Scope

The system calculates monthly electricity bills for residential consumers under two tariff plans: Standard Domestic (tiered slab-based) and Time-of-Use (TOU). It accepts user inputs such as consumer name, consumer ID, consumption figures, and solar export units, then produces a detailed bill breakdown including energy charges, fixed charges, solar credits, and the net payable amount. The application is a console-based Java program and does not include a graphical user interface, database connectivity, or online payment capabilities.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
OOP	Object-Oriented Programming
TOU	Time-of-Use tariff plan with day, peak, and off-

	peak rates
PUCSL	Public Utilities Commission of Sri Lanka
Net Metering	A billing mechanism crediting solar energy exported to the grid
kWh	Kilowatt-hour, the unit of electrical energy consumption
Slab	A consumption range with a specific per-unit rate

Table 1: Definitions, Acronyms, and Abbreviations

2. Overall Description

2.1 Product Perspective

The Modern Residential Electricity Bill Calculator is a standalone, console-based Java application. It does not interface with external databases, web services, or hardware devices. It is designed as an educational project demonstrating Object-Oriented Programming principles such as encapsulation, inheritance, polymorphism, and abstraction. The system models real-world electricity tariff structures and billing logic as defined by the PUCSL.

2.2 Product Functions (Summary)

The system provides the following core functions:

- Accept consumer details (name, ID, solar export units) via console input.
- Allow selection between Standard Domestic and TOU tariff plans.
- Calculate energy charges using the appropriate tariff structure.
- Compute fixed charges based on the selected plan and consumption level.
- Deduct solar energy export credits from the total bill.
- Display a formatted monthly electricity bill summary.

2.3 User Classes and Characteristics

The system serves a single user class: domestic residential consumers or billing clerks who wish to calculate electricity bills. Users are expected to have basic computer literacy, the ability to operate a command-line interface, and access to their monthly consumption figures. No specialised technical knowledge is required.

2.4 Operating Environment

- Platform: Any system with Java Runtime Environment (JRE) 8 or higher.
- Operating Systems: Windows, macOS, or Linux.
- Interface: Command-line console (standard input/output).
- No network connectivity or database is required.

2.5 Constraints

- The application is limited to residential domestic tariffs and does not cover industrial, commercial, or religious institution categories.
- Tariff rates are hard-coded in the TariffRate class; any rate revision requires source code modification and recompilation.
- The system does not persist data between sessions (no file or database storage).
- User interaction is restricted to a text-based console interface.

2.6 Assumptions and Dependencies

- Tariff rates used are those gazetted by the PUCSL effective 15 October 2025.
- Users will provide accurate consumption and solar export values.
- The Java Development Kit (JDK) or JRE version 8 or above is installed on the target machine.
- The system assumes that the solar export rate of Rs. 25.00/kWh remains constant.

3. Functional Requirements

The functional requirements are presented in a structured tabular format. Each requirement is assigned a unique identifier for traceability.

ID	Requirement	Description
FR-01	Consumer Data Input	The system shall prompt the user to enter the consumer's name, consumer ID, and solar export units (kWh).
FR-02	Tariff Plan Selection	The system shall allow the user to select either Standard Domestic or Time-of-Use (TOU) tariff plan.
FR-03	Standard Consumption Input	If the Standard Domestic plan is selected, the system shall prompt for total monthly consumption in kWh.
FR-04	TOU Consumption Input	If the TOU plan is selected, the system shall prompt for day, peak, and off-peak consumption values in kWh.
FR-05	Tiered Energy Charge Calculation	For Standard Domestic consumers, the system shall calculate energy charges using tiered slab rates: Rs. 4.50 (0–30

		units), Rs. 8.00 (31–60 units), Rs. 12.75 (first 60 units if >60), Rs. 18.50 (61–90 units), Rs. 24.00 (above 90 units).
FR-06	TOU Energy Charge Calculation	For TOU consumers, the system shall calculate energy charges as: $(\text{Day} \times \text{Rs. } 35.00) + (\text{Peak} \times \text{Rs. } 67.00) + (\text{Off-Peak} \times \text{Rs. } 21.00)$.
FR-07	Fixed Charge Determination	The system shall apply a fixed charge of Rs. 80.00 for Standard consumers with ≤ 30 units, Rs. 210.00 for Standard consumers with > 30 units, and Rs. 2,100.00 for TOU consumers.
FR-08	Solar Credit Calculation	The system shall calculate solar credit as: $\text{Solar Export Units} \times \text{Rs. } 25.00/\text{kWh}$.
FR-09	Net Bill Computation	The system shall compute the net payable amount as: $(\text{Energy Charge} + \text{Fixed Charge}) - \text{Solar Credit}$.
FR-10	Bill Display	The system shall display a formatted bill summary showing consumer details, tariff type, energy charge, fixed charge, solar credit, and net payable amount.
FR-11	Input Validation	The system shall terminate with an appropriate message if the user enters an invalid tariff plan choice.

Table 2: Functional Requirements

4. Non-Functional Requirements

ID	Category	Description
NFR-01	Performance	The system shall compute and display the bill within 1 second of receiving all inputs.

NFR-02	Usability	The console interface shall provide clear prompts and labels so that a user with basic computer literacy can operate the system without external documentation.
NFR-03	Reliability	The system shall produce mathematically accurate results for all valid input combinations consistent with the defined tariff rates.
NFR-04	Portability	The application shall run on any platform supporting JRE 8 or above without modification.
NFR-05	Maintainability	Tariff rates shall be encapsulated in a dedicated class (TariffRate) to allow rate updates with minimal code changes.
NFR-06	Security	The system does not store or transmit personal data; all inputs are processed in-memory and discarded upon program termination.
NFR-07	Availability	As a local desktop application, the system shall be available whenever the host machine is operational and Java is installed.

Table 3: Non-Functional Requirements

5. External Interface Requirements

5.1 User Interfaces

The system interacts with the user exclusively through a text-based console interface. Input is collected via the Java Scanner class from standard input (System.in), and output is rendered to standard output (System.out) using formatted print statements. The bill output is structured with labelled rows, separator lines, and right-aligned monetary values for readability.

5.2 Hardware Interfaces

The system does not interface with any specialised hardware. It requires only a standard computer with a keyboard for input and a display for output. No peripherals such as printers, barcode scanners, or smart meters are utilised.

5.3 Software Interfaces

The sole software dependency is the Java Runtime Environment (JRE) version 8 or above. The application uses only standard Java libraries (java.util.Scanner) and does not depend on any third-party libraries, frameworks, or external APIs.

6. System Constraints

6.1 Technical Constraints

- The system is implemented in Java and requires the JRE to execute.
- All tariff rates are hard-coded; dynamic rate loading from an external source is not supported.
- The application processes one bill per execution cycle. Batch processing of multiple consumers is not available.
- Input is limited to numeric values for consumption and solar export; non-numeric input may cause runtime exceptions.

6.2 Regulatory Constraints

- The tariff rates conform to the schedule gazetted by the PUCSL effective 15 October 2025. Any regulatory change necessitates a code update.
- The solar export credit rate complies with the current Net Metering policy of Sri Lanka.

6.3 Environmental Constraints

- The application is console-based and does not provide a graphical or web-based interface.
- No data persistence mechanism exists; all data is lost when the program terminates.
- The system is designed for individual use and does not support concurrent multi-user access.

7. Appendices

7.1 Use Case Summary

The following table summarises the primary use cases of the system. A single actor (Consumer/User) interacts with the calculator through the console interface.

UC ID	Use Case	Actor	Description
UC-01	Enter Consumer Details	Consumer	User provides name, ID, and solar export units.
UC-02	Select Tariff Plan	Consumer	User selects Standard

			Domestic or TOU plan.
UC-03	Enter Consumption Data	Consumer	User enters consumption values based on selected plan.
UC-04	View Bill Summary	Consumer	System displays the computed bill with full breakdown.

Table 4: Use Case Summary

7.2 Class Overview

The following table lists the classes and interfaces that comprise the system architecture.

Class / Interface	Description
ElecBillCalculator	Main class containing the entry point; handles user interaction and orchestration.
TariffRate	Data class encapsulating all tariff rates (slab rates, TOU rates, fixed charges, solar export rate).
Consumer (abstract)	Abstract base class defining common consumer attributes (ID, name, solar export units) and abstract billing methods.
StandardConsumer	Subclass of Consumer implementing tiered slab-based energy charge calculation.
TOUConsumer	Subclass of Consumer implementing time-of-use energy charge calculation.
BillingService (interface)	Interface defining the contract for bill calculation and bill printing.
ResidentialBillingEngine	Implements BillingService; aggregates consumer data and tariff rates to compute and print the final bill.

Table 5: Class and Interface Overview

7.3 Glossary

Term	Definition
Domestic Consumer	A residential electricity user billed under the standard household tariff.
Tiered Slab	A consumption bracket with a designated per-unit rate; higher slabs incur higher rates.
Time-of-Use (TOU)	A tariff plan that charges different rates depending

	on the time of day (day, peak, off-peak).
Net Metering	A billing arrangement where consumers with solar panels receive credit for excess electricity exported to the grid.
Fixed Charge	A flat monthly fee added to the bill irrespective of consumption volume.
Energy Charge	The variable portion of the bill calculated from units consumed and the applicable rate.

Table 6: Glossary